

A
Major Project Report
On
**“Early Detection of Parkinson’s Disease Using
Handwriting Images Based on Hybrid CNN-LSTM Model
with Explainable AI”**

Submitted in partial fulfilment of the
Requirements for the award of the degree

IN
MASTER OF COMPUTER APPLICATIONS

Submitted By

Bingi Vivek (24MC201A70)

Under the guidance of
S U Ravi Kumar Chavali
Assistant Professor

DEPARTMENT OF INFORMATION TECHNOLOGY



ANURAG UNIVERSITY

(Approved by AICTE and NBA Accredited)

**Venkatapur (V), Ghatkesar (M), Medchal District, Hyderabad,
Telangana, 500088**

(2024 – 2026)

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**Venkatapur (V), Ghatkesar (M), Medchal District, Hyderabad,
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DEPARTMENT OF INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the project report entitled **“Early Detection of Parkinson’s Disease Using Handwriting Images Based on Hybrid CNN-LSTM Model with Explainable AI”** is a Bonafide work done and submitted by **BINGI VIVEK(24MC201A70)** in partial fulfilment of the requirements for the award of the degree of **MASTER OF COMPUTER APPLICATIONS from Anurag University** during the academic year 2025-2026. The Bonafide work has not been submitted elsewhere for the award of any other degree.

Internal Guide

**S U Ravi Kumar Chavali
Assistant Professor**

Head of the Department

**Dr Niteesha Sharma
Assistant Professor**

External Examiner

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BINGI VIVEK (24MC201A70)

DECLARATION

This is to certify that the major project titled **“Early Detection of Parkinson’s Disease Using Handwriting Images Based on Hybrid CNN-LSTM Model with Explainable AI”** is a Bonafide work done by us in fulfillment of the requirements for the award of the degree **MASTER OF COMPUTER APPLICATIONS** and submitted to **Anurag University, Hyderabad**. I also declare that this project is a result of my own effort and has not been copied or intimated from any source. Citations from any websites are mentioned in the bibliography. This work was not submitted earlier at any other university for the award of any degree.

BINGI VIVEK (24MC201A70)

ABSTRACT

Parkinson's disease (PD) is a progressive neurodegenerative disease affecting the functioning of the brain, mainly movement, balance and writing including handwriting. Timely detection of the disease is crucial for improved management and care of patients suffering from PD. In the present work we propose a deep learning framework for early PD detection based on analysis of handwriting. The proposed detection framework is based on the application of hybrid Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM). The convolutional neural network is used for the feature extraction step of the image. The LSTM is used to model the sequential information that is contained in the image sequences of the images. Specifically, the images of handwriting are segmented into a number of parts for input to the hybrid network.

The Explainable Artificial Intelligence (XAI) techniques based on Grad-CAM are used to improve the interpretability of the models and trust in the decision. The interpretation is performed using visualisation based on Grad-CAM, which highlights the relevant parts of the images that are used by the model to make a prediction. Web-based visualisation is implemented using Streamlit tool for real time prediction of PD detection. The model is trained and evaluated using different performance metrics such as accuracy, F1-score, confusion matrix and classification report with the help of PyTorch framework. The experiment results show the proposed framework is effective for PD detection with handwriting images.

Keywords: Parkinson's Disease Deep Learning CNN-LSTM Handwriting Analysis Explainable AI (XAI) Grad-CAM Medical Image Classification Neural Networks Streamlit Application Early Disease Detection.

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LIST OF ABBREVIATIONS

S.No.	Full Form	Abbreviation
1	Parkinson's Disease	PD
2	Convolutional Neural Network	CNN
3	Long Short-Term Memory	LSTM
4	Hybrid Convolutional Neural Network – Long Short-Term Memory	CNN-LSTM
5	Explainable Artificial Intelligence	XAI
6	Gradient-weighted Class Activation Mapping	Grad-CAM
7	Artificial Intelligence	AI
8	Machine Learning	ML
9	Deep Learning	DL
10	Graphical User Interface	GUI
11	Region of Interest	ROI
12	False Positive	FP
13	False Negative	FN
14	True Positive	TP
15	True Negative	TN
16	Receiver Operating Characteristic	ROC
17	Area Under Curve	AUC
18	Precision, Recall and F1-Score	PRF
19	Portable Network Graphics	PNG
20	Joint Photographic Experts Group	JPEG/JPG

INTRODUCTION

1.1 Overview of The Domain

AI and deep learning make a big breakthrough in a number of areas and one of the most important is healthcare. Neurological diseases such as Parkinson's Disease (PD) are prevalent in a number of patients worldwide and affect their ability to move, coordinate movements, speak and even write. PD is a nervous system disorder caused by reducing the number of dopaminergic neurones affecting movement abilities. The importance of diagnosing PD is the possibility to improve the health state of the patient at an early stage applying a proper treatment.

Traditionally, PD diagnosis was based on clinical examinations, neurological tests and analysis of patients' histories by specialists. But these techniques were often expensive, time consuming and sometimes missed some cases of PD. With the development of technologies and the increasing focus on machine learning algorithms and AI, researchers are taking automated ways to detect Parkinson's disease using handwriting and spiral drawing tests.

The most commonly used algorithms in AI are convolutional neural networks (CNN) and long short-term memory (LSTM) networks. CNNs are very good at extracting visual features from handwriting samples, while LSTMs are useful to analyse sequences in handwriting and drawings. Furthermore, techniques like Explainable AI (XAI) , e.g., Grad-CAM, are able to reveal which parts of the input contributed to model decisions.

The subject area comprises artificial intelligence, medical image recognition, deep learning, and explainable AI.

1.2 Objective of the project

The overall aim of this study is to develop a smart system for early detection of Parkinson's disease by analysing handwriting and spiral images. It would be done by using hybrid CNN-LSTM model in analysis of different features of handwriting. The model will be using the CNN component in extracting various visual cues from images including tremor, shakiness of strokes and other visual cues . The LSTM part will learn the sequential properties of segmented parts of the image patches. One of the objectives of the project would be to improve the understandability of the decision-making processes adopted by the CNN-LSTM

model through the use of the XAI technique of Grad-CAM. Finally, one of the objectives of this project would be to develop a friendly interface for real time detection of Parkinson's disease. This will be achieved by creating a web application using the Streamlit framework.

1.3 Problem Formulation

Parkinson's disease is a degenerative disease of the nervous system with loss of movement functions like writing, movement and coordination. The disease can be difficult to diagnose early because the symptoms may appear gradually and vary. Traditional approaches to the diagnosis of Parkinson's disease are based on expert opinion and observations, which can be expensive, time-consuming, and even inaccurate in early diagnosis. Some of the characteristics of handwriting patterns shown by Parkinson's patients are tremor, jerky stroke and writing instability. This is the reason why the analysis of handwriting images can be an effective technique in determining if someone has Parkinson's disease. However, this process suffers from difficulties due to inconsistencies and hence, an intelligent solution is required for the analysis of the handwriting images automatically.

1.4 Scope

The scope of the work under discussion is confined to the development of an artificial intelligence (AI) system for the diagnosis of Parkinson's disease through handwritten and spiral images. Thus, the main focus of this project is on the areas of image preprocessing, feature selection, pattern recognition, classification and visualisation of results. Deep learning models like CNN and LSTM will be utilised to detect the patterns related to the illness. To improve transparency, we can apply explainable AI techniques such as Grad-CAM to help end users understand the model decisions. Finally, development of a Streamlit application can help users upload their image and get results in a web browser. The described system can serve as an auxiliary tool for research, learning and diagnostics for scientists, students and medical practitioners. However, the scope of the project is limited to image processing and does not replace professional medical diagnosis. This can be improved in the future by having a more comprehensive data set and also identifying other neurological disorders.

1.5 System Requirements

1.5.1 Software Requirements

- Python ≥ 3.8 - a programming language used mainly to train models and

develop applications.

- Pytorch – deep learning framework used to implement the hybrid architecture.
- OpenCV (cv2) – a library used for preprocessing the images like converting to greyscale and resize.
- NumPy - For numerical computation including arrays and tensors.
- Matplotlib – for plotting graphs and visualisation of Grad-CAM results.
- Seaborn – used to plot confusion matrix and evaluation plots.
- Scikit-learn – for evaluation measures of accuracy score, F1 score, confusion matrix and classification report.
- Streamlit - framework to develop the web application interface for predictions and visualisation.
- TensorFlow/Keras – for implementing Grad-CAM Explainable AI visualisation.
- Windows / Linux / macOS ecosystem

1.5.2 Hardware Requirements

- Processor: At least an Intel Core i5 multi-core processor is recommended.
- RAM: 8 GB RAM at least required for model training and testing purposes.
- Disk Space: At least 10 GB of disk space to store datasets, trained models, and other files.
- GPU (Optional): NVIDIA CUDA-enabled GPU highly recommended for faster deep learning training.
- Monitor: Regular display monitor for using the Streamlit-based web application.

2. LITERATURE SURVEY

2.1 Overview of Literature Survey

A literature review is a key component of any research. It helps us to get an insight into past research studies, technologies, research methods as well as current solutions to our problem. Many researchers use artificial intelligence, machine learning and deep learning technologies in their studies on early detection of Parkinson's disease. Traditionally, clinicians used such conventional methods of disease identification as physical examinations, neurologic evaluation and brain imaging scans. To overcome some drawbacks of such methods, the handwriting analysis, spiral drawing and image processing are used for the automatic Parkinson's diagnosis.

Researchers have used various machine learning classifiers such as support vector machine, random forest and decision tree algorithms for the classification of Parkinson's mostly. But in recent years the performance of deep learning algorithms such as Convolutional Neural Networks is even better in visual characteristics extraction [2]. Also, the researchers use Recurrent Neural Networks and Long Short Term Memory network techniques to extract the sequence information from the images. Explainability is performed by means of Explainable Artificial Intelligence techniques, e.g. Grad-CAM technique. According to the literature, it is clear that hybrid deep learning techniques by combining CNN and LSTM achieve more accurate and informative feature extraction results.

2.2 Importance of Literature Survey

The literature review is an important step for studying the existing knowledge and finding possible gaps in the specific field. It presents information on different methods, algorithms, datasets, and evaluation metrics that can be utilised in studies for the detection of Parkinson's disease. The advantages and disadvantages of solutions previously developed can be identified during the review of previous work. Moreover, the literature review helps us to choose appropriate technologies and approaches to create an efficient tool.

Therefore, the literature review of this project provided an opportunity to learn more about the importance of handwriting analysis and deep learning models. The review gave some ideas about CNN and LSTM models that could be used in order to extract useful information.

Besides, during the research process, different types of explainable artificial intelligence such as Grad-CAM were considered.

2.3 Literature Review

2.3.1 U. B. Balci and Others [1] U. B. Balci et al. proposed a deep learning based system to detect Parkinson's Disease using spiral and wave handwriting patterns. The aim of the study was to detect the motor disorders that are reflected in handwriting movements and to improve the accuracy of the disease diagnosis by means of automated analysis.

Method:

The authors used a Convolutional Neural Network (CNN) model to automatically learn and extract meaningful features from handwriting images without the need for manual feature engineering.

Dataset/Features:

The dataset included wave and spiral handwriting samples from healthy people and patients with Parkinson's Disease. The features considered were tremors, irregular curves and stroke distortions.

Results:

Experimental results showed high classification accuracy and successful detection of abnormal handwriting patterns related to Parkinson's disease.

Conclusion:

CNN-based deep learning models are capable of accurately detecting Parkinson's Disease symptoms from handwriting images and offering support for early diagnosis in clinical applications, the study demonstrated.

2.3.2 R. Sharma et al. [2] R. Sharma et al. have proposed an automated framework for the detection of Parkinson's Disease to improve the accuracy and efficiency of the disease diagnosis. The study targeted reducing the human intervention in the feature extraction and improving the classification performance using deep learning.

Methods:

The researchers used a Deep Convolutional Neural Network (Deep CNN) architecture which is able to learn automatically discriminative features from handwriting samples.

Dataset/Features: •

The data set consisted of images of handwriting collected from both healthy individuals and those with Parkinson's. Handwriting irregularities, tremor and stroke distortions were among the features analyzed.

Results:

The model achieved a promising classification accuracy and showed a strong learning ability.

Conclusion

The study concluded that Deep CNN models are effective tools for automated Parkinson's Disease diagnosis and can help medical decision making.

2.3.3. Sivaranjini et al S. [3] S. Sivaranjini et al. conducted a comparative study on the effectiveness of deep learning and traditional machine learning methods for diagnosis of Parkinson's Disease.

Method Applied:

Different machine learning and deep learning algorithms were implemented and compared based on their classification performance.

Dataset/Features:

The dataset contains biomedical and disease related patient information for classification.

Results:

The deep learning models were more accurate and reliable than the traditional machine learning methods.

Summary:

The study showed that deep learning techniques will provide better performance in predicting and diagnosing Parkinson's Disease.

2.3.4 Grover, A., et al. [4] A. Grover et al. proposed a multimodal Parkinson's Disease detection system by combining different patient data sources to improve the prediction performance.

Method:

A CNN-based framework was used for feature extraction and classification in the study.

Dataset/Features:

The input features used were the voice recordings and images of handwriting to predict the disease.

Results:

The integrated approach was more accurate for classification than single-modal systems.

Summary:

The study emphasized the importance of using multiple data modalities for a reliable diagnosis of Parkinson's Disease.

2.3.5 Y. Zhang, et al. [5] Y. Zhang et al. proposed a Hybrid CNN-LSTM architecture for accurate diagnosis of Parkinson's Disease using handwriting analysis.

Methodology:

The CNN part learned spatial image features, and the LSTM part learned sequential dependencies between image parts.

Data set/Features:

The images of handwritten data with Parkinson's Disease features were used for analysis.

Results:

Finally, the hybrid model outperforms standalone CNN models in terms of accuracy and feature representation.

Conclusion:

The study demonstrated that the combination of CNN and LSTM improves the classification performance and the accuracy of disease detection.

2.3.6 M. Rahman et al. [6] Author: Study:

M. Rahman et al. proposed a Parkinson's Disease prediction system based on CNN-LSTM using biomedical signals.

Methods:

We proposed a hybrid CNN-LSTM architecture to learn spatial and temporal information.

Data set/Features :

The model was trained and tested on individual patient biomedical signals.

Results:

The model can extract the features related to the disease and improve the prediction performance.

In summary:

The study demonstrated the usefulness of hybrid deep learning models for biomedical data analysis in identifying diseases.

2.3.7 A. Khan et al. [7] proposed a CNN-LSTM multimodal framework for predicting Parkinson's Disease.

Directions:

This framework applies a combination of CNN and LSTM networks to simultaneously handle multiple feature sources.

Feature / Data set:

This dataset included multimodal data collected from different patients' assessments.

Results:

The model achieved reliable classification results, and improved the accuracy of the prediction of the disease.

Conclusions:

The study demonstrated that multimodal deep learning methods improve robustness and diagnostic performance.

2.3.8 M. Little, et al.[8] M. Little et al. explored speech analysis for the detection and telemonitoring of Parkinson's Disease.

Methodology:

Dysphonia measurements extracted from voice recordings were analysed by machine learning techniques.

Features/Dataset:

Analysis of speech signals and voice features of Parkinson's patients.

Results

The proposed approach was able to effectively classify the disease using speech-based features.

Conclusion:

The study demonstrated that voice analysis could serve as a reliable and non-invasive diagnostic tool.

2.3.9 B.E. Sakar and others.[9] B. E. Sakar, et al. Comparative study of speech signal processing algorithms for classification of parkinsons disease.

Method Employed:

Several machine learning models and feature extraction techniques were experimented with.

Feature/Dataset:

We used speech recordings of healthy individuals and of Parkinson's patients.

Results:

Several algorithms achieved good classification performance.

In summary:

The study underscored the effectiveness of speech processing techniques in the diagnosis of Parkinson's Disease.

2.3.10 Tsanas A. Et al. [10] A. Tsanas et al. presented techniques of speech signal processing for the classification of Parkinson's Disease.

Method of preparation:

They used speech data and new feature extraction and machine learning techniques.

Dataset/Features Dataset:

They studied voice recordings of speech with voice impairments related to the disease.

Results:

The algorithms they proposed could achieve very high classification accuracies.

Summary:

The study showed that speech analysis techniques can improve the performance of the disease detection significantly.

2.3.11 R. Prashanth et al. [11] R. Prashanth et al. developed a machine learning framework for Parkinson's Disease detection using multimodal features .

Methodology:

Different machine learning algorithms were used for disease classification.

Dataset/Features:

The dataset contains a number of biomedical and clinical features.

Results:

The multimodal approach enhanced the accuracy of classification and the reliability of prediction.

Summary:

The study confirmed that by combining multiple types of features the diagnosis of Parkinson's Disease can be improved.

2.3.12 C. Salvatore et al [12] C. Salvatore, et al. The research on the application of machine learning techniques for the diagnosis of Parkinson's disease using Brain MR Images. 9. This was to help find the structural characteristics of brain related to PD and utilised for classifying the PD successfully .

Methods:

ML techniques are used to extract the MR Image based features of brain images. The efficiency of distinguishing patients suffering from PD from normal people was studied by trying a number of classification methods in the experiment.

Dataset/Features:

Experiment is performed on dataset of MR Images of PD patients and healthy subjects. Brain structural characteristics and images for feature extraction.

Results:

The built ML model showed satisfactory performance in the classification of Parkinson's disease patients.

Summary:

Machine learning approaches using the MR Images can be used as reliable tools for diagnosis and help physicians to make a decision during diagnosis.

2.3.13 P. Drotar, et al. [13] The kinematic and pressure features of handwriting were analyzed for the purpose of diagnosis of Parkinson's Disease by P. Drotar et al. The purpose of this study was to look for patterns and abnormalities in the handwriting of Parkinson's patients that appear commonly due to motor deficiencies.

Methodology:

Machine learning was used to analyze the dynamics of the handwriting. The subjects were classified based on these handwriting features extracted. Statistics and computational methods were used to extract and classify the features of handwriting.

Features/Dataset:

The dataset collected handwriting samples of normal and Parkinson's Disease individuals. The features extracted included speed of writing, pen pressure, and time of stroke. Also analyzed were movement paths and writing stability.

Results:

The features obtained from analyzing handwriting were found to be discriminating of Parkinson's Disease and non-Parkinson's Disease subjects with encouraging classification rates.

Summary:

The findings indicated that Handwriting features provide a suitable biomarker for initial detection and diagnosis of Parkinson's Disease through non-invasive methods.

2.3.14 C. Pereira et al. [14] Parkinson's Disease Classification by Machine Learning Techniques on Handwriting Dynamics. C. Pereira et al. authors studied the possibility of using movement-related handwriting features to classify Parkinson's Disease.

The methodology used:

After building classifiers, handwriting features were extracted and performance of different machine learning classifiers in discriminating control patients from PD patients was evaluated.

Data Set/Features:

Data consist of dynamic handwriting data, i.e. handwriting velocity, acceleration, stroke pattern and motion behaviour while writing.

Results:

The proposed ML based approach showed promising classification accuracy and utility of handwriting dynamics for disease detection.

Conclusion:

Handwriting is a low-cost, easy and effective technique to detect PD and could supplement early clinical assessment.

2.3.15 B. Eskofier et al. [15] B. Eskofier et al. reviewed recent advances in machine learning for the monitoring and assessment of Parkinson's Disease with wearable sensors and mobility analysis systems. The study involved continuous monitoring of patients' movements and symptom development.

Method Used:

Machine learning algorithms and sensor-based analytical techniques were studied to evaluate their efficiency in the identification of Parkinson's Disease symptoms and mobility impairments.

Dataset/Features:

The reviewed studies used data from wearable sensors, such as gait patterns, walking speed, body movement information, and motion sensor measurements collected from patients with Parkinson's disease.

Results:

The methods analysed show a great potential for continuous disease monitoring, symptom prediction and patient evaluation in real-life environments.

In summary:

The study emphasised the increasing significance of wearable technologies and machine learning methods for Parkinson's Disease management, early diagnosis and long-term patient monitoring.

Table 2.3.1: Functional Comparison of Existing Systems

S.No	Author & Year	Approach	Parameters Used	Technique / Algorithm	Advantages	Limitations
1	Balci et al. (2019)	Parkinson's detection using spiral and wave drawings	Spiral & Wave Images	CNN-Based Deep Learning	Effective feature extraction and classification	Requires large dataset
2	Sharma et al. (2020)	Automated Parkinson's Disease Detection	Handwriting Images	Deep CNN	Automatic feature learning and high accuracy	Limited generalization
3	Sivaranjini et al. (2019)	Parkinson's Diagnosis using Deep Learning	Biomedical Data	Deep Learning Model	Improved classification performance	Requires parameter tuning
4	Grover et al. (2021)	Parkinson's Detection using Voice and Image Data	Voice & Image Data	CNN	Combines multiple data modalities	Complex architecture
5	Zhang et al. (2021)	Parkinson's Disease Diagnosis	Sequential Data	Hybrid CNN-LSTM	Captures spatial and temporal features	High computational cost
6	Rahman et al. (2022)	Parkinson's Detection using Biomedical Signals	Biomedical Signals	CNN-LSTM	Effective temporal feature learning	Requires large training data
7	Khan et al. (2023)	Multimodal Parkinson's Prediction	Multimodal Dataset	Multimodal CNN-LSTM	High robustness and accuracy	Complex implementation
8	Little et	Telemonitori	Voice Dataset	Speech Signal	Suitable for	Sensitive to noise

	al. (2009)	ng of Parkinson's Disease		Analysis	remote monitoring	
9	Sakar et al. (2013)	Speech- based Parkinson's Classificatio n	Voice Dataset	Speech Processing Algorithms	Effective speech feature extraction	Performance depends on data quality
10	Tsanas et al. (2012)	High Accuracy Parkinson	Voice Dataset	Advanced Speech Processing	High classification accuracy	Computational complexity

2.4 Research Gaps

Although there are many studies available for Parkinson's disease detection, there are still many challenges with the current approaches. First, most of the traditional machine learning approaches rely on hand-crafted feature selection algorithms that require domain-specific knowledge and are inefficient in detecting handwriting complex features. Secondly, some research models treat the process as an image classification one without looking at the sequential aspect of handwriting. This approach reduces the efficiency and the overall accuracy of the disease detection systems. Furthermore, the dataset used in many studies is not rich enough as it is quite small and imbalanced.

Another research problem is the non-interpretability of deep learning algorithms. The existing models can predict the class but they cannot show why they classified the stroke in this way. In this way, the lack of explainability prevents the healthcare workers to use such systems since they cannot be sure about the results generated by such approaches. Furthermore, a number of existing solutions do not provide real-time analysis and intuitive user interface, which makes them not convenient for use in practice. Finally, some approaches assume the use of expensive medical devices which require complex preprocessing steps.

2.5 Project Objectives

The aim of this project is to develop an intelligent machine learning model by deep learning for early disease detection. In this sense, the main purpose of this project is to improve the efficiency of disease detection through the analysis of the mentioned symptoms. The patterns include tremor, unsteady handwriting and other problems to do with irregularities of the hands. This is the reason a hybrid CNN-LSTM model will be implemented. In particular, the

CNN function is to extract the visual data from the given images, while the LSTM will detect the relations between the various image patches.

In addition, the interpretability of the predictions should be improved through explainable artificial intelligence techniques such as Grad-CAM. Another is that this project will be performance-driven. It is done by the help of effective pre-processing, training and testing techniques. It is also possible to estimate the performance of models using various performance metrics such as accuracy, F1-Score, confusion matrix etc. The additional goal is to develop a user-friendly web application with the use of Streamlit technology.

2.6 Project Justification

Parkinson's disease (PD) is one of the most prevalent neurological diseases worldwide and the number of patients is increasing. This disease slowly starts to affect the patient's motor skills, coordination, speech and handwriting. Early detection of the disease is very important as the symptoms of patients can be controlled with the medical treatment and therapy. Traditional methods for disease detection are largely dependent on doctor observation and neurological examinations. Such diagnosis is not always simple and cost-effective, especially in the earlier stages when it is necessary to diagnose the disease before its symptoms become more apparent.

Recent advances in the field of AI and deep learning have provided a possibility to design automatic medical disease detection algorithms. One of such algorithms is based on image processing and handwriting analysis. Strange stroke shapes, irregular hand writing and trembling are usually indications of Parkinson's illness. Therefore, the process of handwriting images can be used for early detection of diseases. Our algorithm is a combination of CNN and LSTM neural network to process handwritten pictures. The combination was chosen because CNN is appropriate for image processing and LSTM can extract sequential data from handwriting. In addition, Grad-CAM allows the model to provide some additional visual information about the explanation of its results. The development of an app based on Streamlit makes the system user-friendly and easy to use.

3. METHODOLOGY

3.1 Introduction

Methodology is an important element that provides detailed information about the process of development of the proposed system, techniques, algorithms and procedures that take place in the process. A hybrid deep learning approach for the detection of Parkinson's disease based on handwriting and spiral images. In particular, the proposed methodology will mainly focus on handwriting abnormalities, which are characteristic for this disease, including tremors, shaky stroke, irregular curves and other factors.

Parkinson's disease is a chronic, progressive disorder that adversely affects the patient's movement, coordination, speech and ability to write. It is very important to diagnose this disease early, in order to provide the right medical treatment and improve the patient's quality of life. The main problem in the diagnosis of Parkinson's disease is the inability of specialists to correctly diagnose at an early stage when the symptoms of this disease may be hardly noticeable. Moreover, manual checking of handwriting abnormalities is a very difficult and time-consuming task because it is difficult for doctors and medical experts to notice the tremors and irregular handwriting pattern. Therefore, an intelligent Parkinson's disease detection system is needed.

The main motivation to employ a deep learning technique is the potential to extract meaningful features and analyse images without the need of manual feature extraction. Machine learning techniques can hardly detect all complex features as specialists should manually find hidden handwriting features.

The process will start with preprocessing stage where handwriting images will be converted into greyscale images. They will also be normalised and cropped as image patches. Then, the CNN network will be used to recognise the irregularity of the spatial handwriting such as tremors and shaky strokes. We will use LSTM to find the sequential relation between the different patches of the image. Moreover, the usage of Grad-CAM Explainable AI technique will provide visualisations, which will assist experts to underline critical areas that greatly influence the prediction outcome. Finally, the development of the Web App will be carried out with the use of Streamlit.

Therefore, the proposed methodology will be a combination of different techniques and approaches related to deep learning and visualisations, which makes the system highly accurate, efficient, reliable and interpretable.

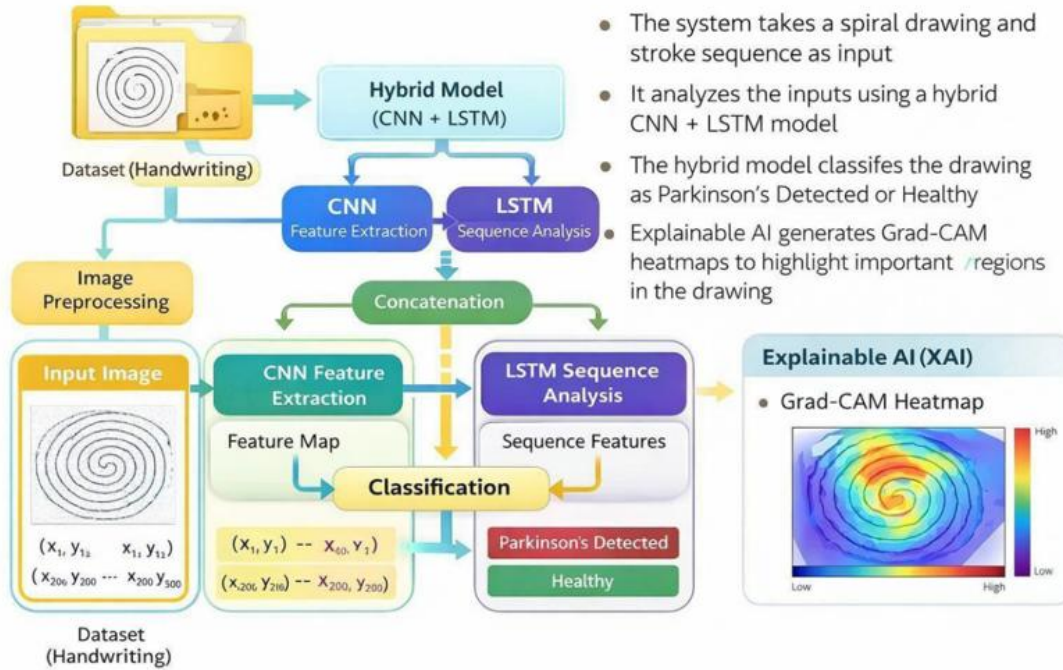


Fig. 3.1.1 System Architecture of "Early Detection of Parkinson's Disease Using Handwriting Images Based on Hybrid CNN-LSTM Model with Explainable AI"

3.2 Proposed Methodology

The proposed Parkinson's disease detection system is for detecting Parkinson's disease using handwriting and spiral images via deep learning models. The system analyzes the samples of handwriting and spiral drawings and classifies whether the given patient suffers from Parkinson's disease or he is healthy. This system provides the healthcare professionals a quick, reliable and automated way of screening out potential patients suffering from Parkinson's disease. This is a lightweight, fast and efficient system for real time prediction applications. The classification of the disease is done with the consideration of spatial and sequential pattern recognition using a hybrid CNN-LSTM.

1. Data Collection

The data collection is the handwriting, spiral images of the patients suffering from the Parkinson's disease and healthy individuals. The data collection consists of different handwriting patterns and spiral images with different writing qualities. Proper labeling of images so that deep learning model learn and predict the disease correctly.

2. Image Preprocessing

After collection, the images are pre-processed to improve their quality and consistency. Image preprocessing steps include grayscaling, resizing, normalization, and noise reduction. The size of the images are resized into standard size and pixel values are scaled . This pre-processing step reduces computational complexity and improves model accuracy.

3. Hybrid CNN-LSTM Classification Module

CNN-LSTM network is given preprocessed images. The images are split into several patches of sequences. Then these sequence patches are passed to CNN layer. The CNN extracts meaningful visual features of the image which includes tremors, shaky lines, distorted curve etc. The feature maps are then inputted to the LSTM network. The LSTM learns the sequence relations and handwriting pattern from the image. Then, in this case, the input image is classified as healthy or Parkinson's disease.

4. Feature Analysis and Prediction

The predictions are made using the trained CNN-LSTM model on extracted handwriting features. Then the features are further analyzed to predict whether it belongs to a healthy person or a patient suffering from parkinson disease. It provides confidence score for the prediction.

5. Visualisation and Explainable AI

The heatmap was generated using the Grad-CAM algorithm to explain which part of the image the model is focusing on for prediction.

6. Output and Result Presentation

Finally, the system outputs the prediction, confidence score and Grad-CAM heatmap with a Streamlit based interface.

3.3 Proposed Algorithms

Algorithm: Algorithm for detecting Parkinson's disease

Spiral & Handwriting image database

Image Size (28x28)

Sequence Length=4

No. of Epochs Batch Size

CNN-LSTM Model Parameters Learning rate

The Ministry of Public Works and Transport will be launching a new tender for the construction of the new bridge over the river Tormes in Salamanca. The new bridge, to be constructed by the cantilever method, will have two lanes, one in each direction. The project is estimated to cost some 25 million euros and will take about two years to complete. The old bridge is in a bad way and can no longer cope with the volume of traffic so it will be replaced by the new bridge.

Parkinson's Disease Classification (Healthy / Parkinson's Disease) Confidence Score Grad-CAM Heatmap Visualization START

Step 1: Import Required Libraries

- PyTorch
- OpenCV
- numpy library
- Matplotlib.
- Seaborn (for visualization)
- scikit-learn
- Streamlit

Step 2: Load Handwritten Image Data and Spiral Data

- Load all Parkinson's and healthy images
- Categorization of the images to respective classes

Step 3: Image Pre-processing

- Convert images to grayscale

- Resize images to 28x28 pixels.
- Normalize pixel value to range 0 to 1
- Remove unwanted noise• Divide each image into a sequence of patches

4. Split the data into Training Data Set

- Data for validation
- Testing Data

Step 5: Develop Hybrid CNN-LSTM Model CNN Module CNN Module

- Extract spatial feature from patches of image
- Detect tremors, shakiness, irregular lines

LSTM module 2.3

- Find the sequence relation between image patches
- Preserve the flow of the handwriting and the continuity of the stroke

Step 6: Model Parameters Learning rate

- Batchsize
- eras
- Adam-Optimizer
- Loss Function - Cross Entropy

Step 7: Training Phase

FOR each epoch DO FOR All Batches

- CNN-LSTM model with input of sequences of images
- CNN for extracting deep visual features
- Learn sequences using LSTM
- Generate classification output
- Predict class label - Healthy - Parkinson Disease
- Calculation of Cross Entropy Loss
- Adam Optimizer for updating model weights

END FOR

* Compute Validation Accuracy

- Compute Validation Loss

END FOR

Step 8: Save the Best Trained Model

IF Highest Validation Accuracy

- Saving the trained model CNN-LSTM

IF Test Step 9:

- Load test picture • Preprocess
- Use trained model to make predictions
- Calculate confidence score

Step 10: Develop Explainable AI Visualization

- Using the Grad-CAM algorithm
- Generate Heatmap
- Mark important areas of handwriting that impact prediction

Step 11: Evaluate model performance

- Precise • Correctness
- In Memoriam • F1 Punktzahl
- Classification of Reports • Confusion Matrix

Step 12: Show the end result

- Predicted Result (Parkinson's / Healthy)
- Confidence level
- Visualizing Grad-CAM Heatmap

3.4 Methodology Design

The methodology design of the proposed system for Parkinson's Disease detection is based on a structured workflow that combines Deep Learning and Explainable Artificial Intelligence (XAI) techniques. The process starts with the recording of the handwriting or spiral drawings of the users and dataset collection for analysis. The hybrid Deep Learning model takes the collected images as input.

The input images are preprocessed to enhance the image quality and to make them ready for analysis. The processed images then are fed to the Hybrid CNN-LSTM. The Convolutional Neural Network (CNN) extracts salient visual features such as tremor, shaky strokes, irregular curves, and handwriting instability. Simultaneously, the Long Short-Term Memory

(LSTM) network analyses the sequential relations and the flow of the handwriting by learning the temporal dependencies present in the image patches.

A feature concatenation process is used to combine the CNN extracted features with the sequence information learned by LSTM. This step merges the spatial and sequential information to form a holistic representation of the handwriting pattern. Then the classification module classifies if the handwriting belongs to a Healthy individual or a Parkinson's Disease patient based on the combined features.

The system uses Explainable Artificial Intelligence by means of Grad-CAM algorithm in terms of transparency and interpretability. The Grad-CAM produces heatmaps that highlight the important regions of the handwriting image that contribute the most to the model's prediction. These visualizations help the user and healthcare professional understand why the model made a certain classification result.

Finally, the system shows the prediction results with the Grad-CAM visualisation and saves the results for future use and analysis.

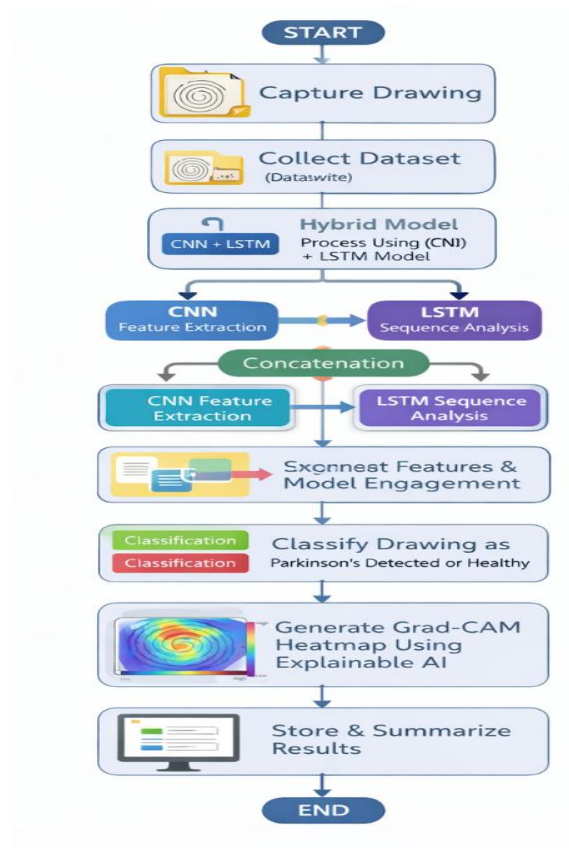


Fig. 3.4.1 Flowchart of Proposed Methodology for Early Detection of Parkinson's Disease Using Handwriting Images Based on Hybrid CNN-LSTM Model with Explainable AI

The complete workflow of the proposed Hybrid CNN-LSTM based Parkinson's Disease Detection System with Explainable AI (Grad-CAM) is shown in the following flowchart. It begins with the acceptance of the handwriting samples and ends with the classification of diseases and its output.

Step 1. Sketching

The user writes a draw a spiral or handwriting and sends for the processing. These drawings contain the fundamental motor symptoms of PD.

Step 2. Data Collection

The draws are collected, saved and added to the healthy and PD datasets. The model is trained and tested on the combined data set later.

Step 3. Hybrid CNN-LSTM Model

The collected image will be feed to the CNN-LSTM Hybrid model which is the combination of two different types of networks CNN and LSTM in order to get the benefit of pros of both.

4. CNN based Feature Extraction

The CNN model is used to extract the relevant visual feature of handwriting images like tremor, irregular curved shape, distorted handwriting strokes, shaking writing patterns etc.

Step 5. Sequence Analysis using LSTM

LSTM model will analyse the sequences and temporal relations among different segments of an image, thus they can predict writing patterns well.

Step 6. Concatenation of features

The features extracted by CNN and LSTM will be concatenating to produce more sophisticated feature with both spatial and sequential information.

Step 7. Fusion of features and inference of model

The combined features will be fed into the classification layers to learn specific PD patterns. This step further improves the classification accuracy of the system.

Step 8. Order

The trained model will recognise the handwriting as one of the following classes: PD Healthy Detected

Step 9. Generate Grad-CAM Heatmaps

The system is made transparent by using Grad-CAM to visualise the regions in handwriting images which have high influence on the system decision making process.

Step 10. Data Storage & Report

The prediction and classification outputs as well as the Grad-CAM results will be saved for further analysis.

Step 11. Finished

The system will end with the last prediction and explainability.

4. Implementation

4.1 Introduction

Implementation is the stage in which the proposed Parkinson's Disease detection system is converted from a conceptual design into a working application. In this phase, the Hybrid CNN-LSTM model is designed, trained, tested and deployed for handwriting and spiral images analysis. The implementation process consists of: dataset preparation, image preprocessing, feature extraction, sequence learning, classification, evaluation and result visualization.

The proposed system is developed using Python programming language and various Deep Learning libraries like PyTorch, OpenCV, NumPy, Scikit-learn, Matplotlib, Seaborn and Streamlit . The CNN component of the model extracts relevant visual features from the images of handwriting, while the LSTM component learns sequential dependencies between patches of the image. The combination of these two models improves accuracy and reliability of Parkinson Disease detection.

During the implementation, the dataset is processed by different pre-processing techniques like grayscale conversion, image resizing, normalization and sequence generation. The processed data is further used for training and validation of the Hybrid CNN-LSTM model. The model is trained using Adam optimizer and Cross-Entropy Loss function to improve the classification performance.

Explainable Artificial Intelligence (XAI) techniques are used with Grad-CAM to improve transparency and interpretability of the system. The method produces heatmaps, indicating important regions in the handwriting image that contribute to the model's prediction. Finally, a web-application based on Streamlit is developed t

provide real-time image upload, prediction results, confidence scores and visualisations. The implementation phase ensures efficient operation of the proposed system and accurate, reliable and explainable detection of Parkinson's Disease using handwriting images.

4.2 Implementation

Implemented Hybrid CNN-LSTM Deep Learning Model for Proposed Parkinson's Disease Detection System. The system is developed in Python language and uses several libraries such as PyTorch, OpenCV, NumPy, Scikit-learn, Matplotlib and Streamlit. The dataset consists of two classes of images, Healthy and Parkinson's Disease with handwriting and spiral images.

Images are first pre-processed using a number of different operations including conversion to grey-scale, resizing, normalisation and sequence generation. The images are divided into patches and serve as an input sequence to the model. The CNN part learns the most important visual features, such as tremors, shakiness of strokes, irregularity of curves, instability of handwriting, etc. The LSTM network takes the CNN extracted features as input and learns the sequential pattern and temporal dependencies between the image patches.

The model is trained with Adam optimiser and Cross Entropy Loss function for better classification purposes. We train the model on the training datasets and evaluated on validation sets, to judge model performance and avoid overfitting. We save the best performing model for predicting the images.

The system employs the Grad-CAM visualisation for enhancing transparency and interpretability. It produces heat maps indicating which parts of the image contributed to the prediction result. Finally, a web application was developed using Streamlit to allow users to upload images of handwriting and receive predictions, confidence scores and Grad-CAM visualisations in real-time.

4.3 Experimental Setup

Experimental setup, hardware and software used to design, develop, train, test and evaluate the performance of the proposed Parkinson's Disease detection system. The developed system is in programming language Python. The developed system uses many libraries like Deep Learning, image processing and image visualization libraries. The experiments performed evaluate the capability of the Hybrid CNN-LSTM system to classify the handwriting images into Healthy and Parkinson's Disease classes. The hardware specification used to design the project is Intel Core i5 or higher, 8 RAM, storage for the data set and trained models. Optional use of NVIDIA GPU for faster model training and computational acceleration (CUDA supported). Software specification used: Python 3.8+, Pytorch, OpenCV,

NumPy, Scikit-learn, Matplotlib, Seaborn, Streamlit. The dataset is classified into two groups: Healthy and Parkinson's Disease, with images of handwriting and spirals. The training dataset is 80 percent of the whole dataset and the validation dataset is 20 per cent. The Hybrid CNN-LSTM system is trained using the training dataset and evaluated using the validation dataset. The optimiser used is Adam and the loss function used during training is Cross-Entropy Loss. The effectiveness of the proposed system will be demonstrated using different evaluation parameters such as Accuracy, F1-Score, Confusion Matrix, Classification Report, etc. Heat maps for the images will be generated using Grad-CAM visualisation which will help to find out which is the relevant part in the input images which affects the output of the system.

4.4 Step-by-Step Implementation

Step 1: Import Libraries

Import of necessary Python libraries and frameworks: Before we start to build a system, first we start by importing of necessary Python libraries and frameworks to help us build the system. Each of the libraries would give us the capabilities to do Image Processing, Deep Learning, Model Evaluation and Visualisation. We will build and train a Hybrid CNN-LSTM model using PyTorch, one of the most popular and versatile open-source machine learning frameworks. Basic image processing functions like grey scaling of images, Resizing images and Loading of images are done using the OpenCV library. We'll need NumPy for basic numerical and mathematical operations. It is also used to perform a wide variety of arithmetic operations. NumPy provides objects like multidimensional arrays and related fast operations on arrays. Scikit-learn library has metrics like accuracy score, F1 score, confusion matrix, classification report, etc., to measure the performance of a model. For plotting graphs of training accuracy curves, loss graphs and also the confusion matrix, the Matplotlib library would be used. The Seaborn library, which is an open-source Python data visualisation library based on Matplotlib, is also used to plot and visualise data graphically. Streamlit is also an open-source Python library that helps in building web applications.

Step 2: Load Datasets

In the 2nd step, the dataset that has been selected for the training and testing of the handwriting and spiral images is loaded. It is then decided to make 2 classes of the dataset to label them with 'Healthy' and 'Parkinson's disease '. In these two classes, images of handwriting belonging to the respective class will be considered. A custom dataset class is

written that will read image files from the dataset directory and then label the image according to its class. The paths of the image are saved and stored. It is decided that images will be loaded from the grayscale form to reduce the computational cost and effort. Data is loaded into tensor data, which is required by the DL models from image files by the dataset loader. Careful loading of the data should be undertaken so that the model performs best. In this stage, all images and their class labels are loaded and stored for image processing and training.

Step 3: Handle Missing Values

Before training the model data is thoroughly examined to avoid missing, corrupt or unreadable image files. Low quality or corrupted images can degrade the performance of a model and even cause errors during training. Validation checks are performed to find bad images in the dataset and images that are not readable by OpenCV are excluded from training. All image files are verified to have a valid category and have some actual image data. Verifying that an image file contains meaningful image data makes sure that the files aren't empty, or just full of null values. The validation step makes sure that the training data is 'clean' images that the model can work with; a lot of computation is spent on ensuring that the model has useful data to learn from. Detecting and handling invalid data helps prevent unexpected run-time errors and improve model stability, removing bad samples for clean and consistent data will increase classifier accuracy.

Step 4: Train-Test Split and Model Training

The data is split into a training and validation set in the ratio 80:20. In the training phase, a Hybrid CNN-LSTM is trained with the training set and the performance of the trained model on the validation set is evaluated. The CNN part of the model is utilized for extracting visual features like tremors, shake strokes and crooked lines from the images of writing. The LSTM part of the model takes in these features and learns the relation between the image patches in a sequential manner. The Adam optimizer and Cross-Entropy Loss are used to train the model. It does forward propagation, calculates the loss, back-propagates and updates the weights in each epoch. The learning rate scheduling and early stopping is used to prevent the over-fitting of model. Training continues until the model performs at its best. We save the trained model with the best performance and use it for evaluation and prediction.

Step 5: Data Visualisation

It includes visual depiction of performance of the proposed system to identify working and performance of Parkinson's Disease detection system. During and after training of a model you get different plots and visualizations. The process of learning is understood by plotting loss of training set and loss of validation set. Also, over-fitting/under-fitting of model is checked. The accuracy plot at each epoch is to measure the enhancement in prediction accuracy over each epoch. The confusion matrix plot shows relation between Predicted and Actual class. Classification report is used to describe accuracy of the model with all parameters like precision, recall, F1 score, accuracy. Heatmaps of Grad-CAM are generated to create a part of Explainable AI component. They show the more informative parts to model for its prediction in handwriting image.

5. Performance Evaluation

5.1 Introduction

The performance evaluation phase is carried out to verify the efficiency and reliability of the Hybrid CNN-LSTM model for the detection of Parkinson's Disease from handwriting and spiral images. It measures the performance of the proposed model in predicting Parkinson's disease from the normal cases. Moreover, the advantages and disadvantages of the proposed method are presented.

To evaluate the model, unseen test samples were given to the trained model and compared with the actual class of the predictions. The classification performance is evaluated using different metrics and the efficiency of the model is analyzed. These matrices provide a clear mathematical picture on the efficiency of the model to distinguish Parkinson's disease handwriting.

Not only mathematical estimation, visualization with Confusion Matrix and ROC Curve were analyzed to see behavior of model and XAI with Grad-CAM was adopted to recognize those regions that affect the prediction of classification. The results show that the proposed system has a high potential to provide accurate, reliable and interpretable prediction.

5.2 Performance Metrics Used

The proposed model for the detection of Parkinson's Disease is evaluated with the help of different performance measures. The classification accuracy of the classification system can be evaluated by the performance measures of the classification system.

Accuracy It measures the accuracy of the model. It is the ratio of number of correctly classified samples to the total number of sample cases.

Precision: It tells accuracy of the classification of positive samples with respect to all predicted positive samples. It measures how accurate it is in correctly identifying the samples diagnosed with Parkinson's.

Recall: This defines the ability of the classification model in predicting positive cases for the samples with Parkinson's, it measures how sensitive it is for correctly predicting a positive class.

F1-Score: It is the harmonic mean of the recall and precision value and is used for unbalanced classification.

AUC (ROC) : It is called as Area Under the Receiver Operating Characteristic Curve and provides measure on the accuracy of the classifier in correctly classifying either the healthy cases or the cases of Parkinson's Disease.

Confusion Matrix: This is useful for visualizing the outcome of a classification to see where each instance falls within each class.

Accuracy:

Accuracy The performance metric which defines the overall performance of the proposed model. This is the percentage of misclassified samples with respect to the total number of test samples. The higher the accuracy value, the better the classification between PD and Healthy patients.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision:

Precision is the ratio of actual Parkinson's Disease cases that the model predicts as Parkinson's Disease. This tells us how correct the positive predictions are. This is a measure of how much the model commits false positives. A model performs robustly and achieves high precision results.

$$\text{Precision} = (\text{TP} / (\text{TP} + \text{FP}))$$

Recall:

Recall measures the ability of the model to correctly predict positive real samples of Parkinson's Disease in the dataset. It is expressed as the total number of true positive predictions divided by the total number of positive samples. The higher recall indicates that the model is able to detect most of the Parkinson's Disease samples and only few samples are missed.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1-Score:

F1-Score is the harmonic average of Precision and Recall. It is a model based on number of false positives and false negatives so the accuracy of the model can be measured by balancing both. This model is used for unbalanced data sets. It measures the classification performance.

$$\text{F1-Score} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Confusion Matrix:

The confusion matrix is used to evaluate the performance of classification. This is a table showing the number of correct and incorrect classifications the model has made for each class. It has true positives, true negatives, false positives and false negatives. It is useful when studying the pros and cons of the proposed model.

5.3 Results and Analysis

The proposed Hybrid CNN-LSTM was tested with Handwriting and Spiral images datasets for diagnosis of Parkinson's disease. The experimentation result proved that the developed model could effectively extract the spatial and sequential features from handwriting pattern images. CNN extracts the significant visual features like tremor, uneven curve, shaky handwriting stroke, and shaky characteristic of handwriting image and LSTM extracts sequential relationship, spatial relationship, and time relation between image blocks, which enhance the classification accuracy.

The performance analysis of the system was measured using some evaluation parameters like accuracy, precision, recall, f1-score, AUC (ROC) and confusion matrix. This evaluation measures the capability of system for classifying the healthy handwriting and Parkinson handwriting image samples and the values shows that the system can properly classify the handwriting images and can produce accurate prediction results. The confusion matrix measures the number of correctly and incorrectly classified samples for thorough and effective assessment of system.

The proposed Hybrid CNN-LSTM is also explained using visualization method called Grad-CAM to provide heat maps on the regions which greatly affect the system for classification so that the system becomes explainable and clear to humans. In general, the analysis proved the effectiveness and reliability of the developed model and its suitability for Parkinson disease detection.

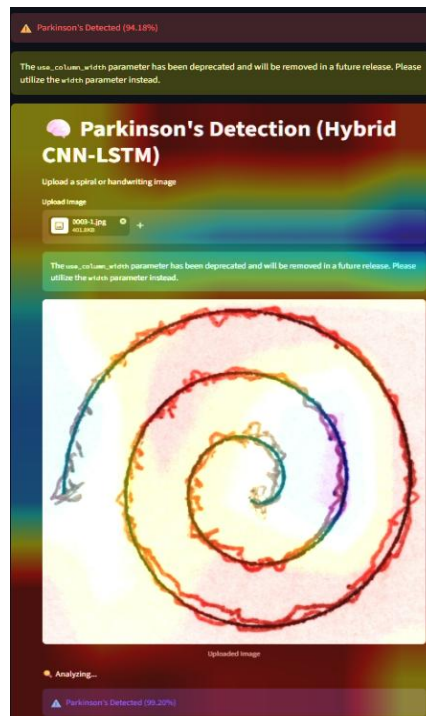


Fig 5.3.1 Grad-CAM Visualisation Output Highlighting Regions Influencing Parkinson's disease

Disease Prediction

Disease prediction: prediction of whether the patient has the disease using some health related information, such as symptoms, image, biomedical data etc. The prediction system of disease is becoming more efficient and precise now with the development of the AI. The system helps physicians to diagnose the disease at early stage, and benefits patient by having early treatment.

In the task of Parkinson's disease prediction, there are some kinds of data such as handwriting image, speech signal, biomedical data, movement pattern to recognise those signals which reflects symptoms of disease to diagnose whether the patients has disease or not. The introduction of deep learning technique allows the automatic learning of the important feature in the data with CNN and LSTM model which leads to the better accuracy of disease prediction. CNN-LSTM combined features of spatial and temporal extraction can do well in disease prediction tasks. Accurate disease prediction can reduce the misdiagnosis rate and provide better help in clinical treatment.

5.3.1 Training and Validation Accuracy Graph

Training & Validation accuracy graph- this is a necessary graph in terms of studying learning behaviour of proposed Hybrid CNN-LSTM model during its training phase. The y-axis represents the accuracy values and the x-axis represents the epoch number for training and validation set. Training accuracy is how well a model learns and predicts features of a data-set unseen in the training set. The validation accuracy is the accuracy with which the model predicts a new unknown data-set.

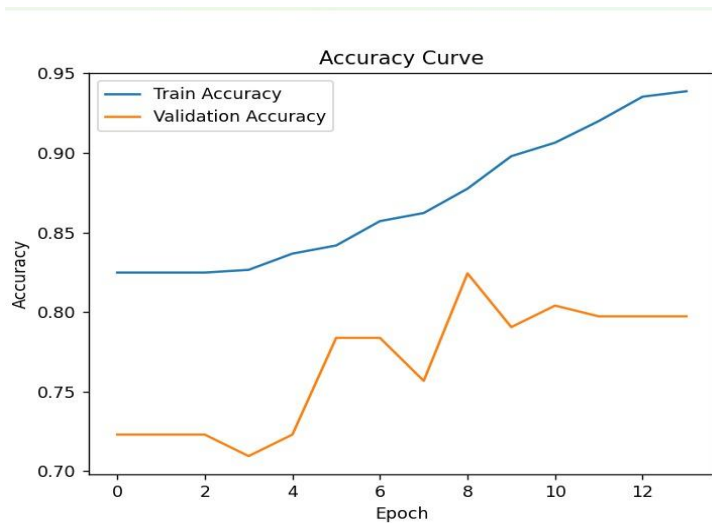


Fig 5.3.2 Training and Validation Accuracy Graph

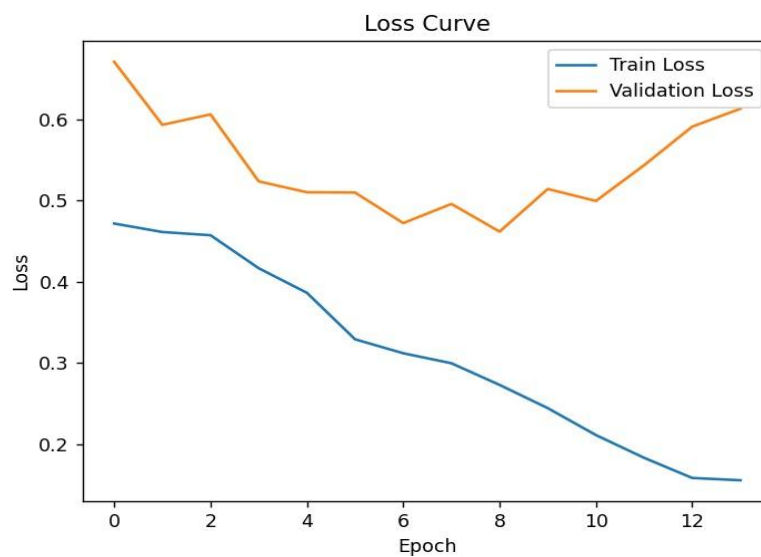


Fig 5.3.3 Training and Validation Loss Graph

The increase in training and validation accuracy shows that the model is learning and predicting the output for the Parkinson's Disease properly. A high training accuracy with compared to the validation accuracy can be an indication of over-fitting, where the model may learn the data-set instead of its features. Low values mean under-fitting. Researchers examine the graph of training and validation accuracy to study the learning characteristics of model, to diagnose a problem during the training and to fine tune the hyper-parameters. In this study, the training and validation accuracy of the proposed model is taken to prove that the model learns and predicts Parkinson's Disease correctly.

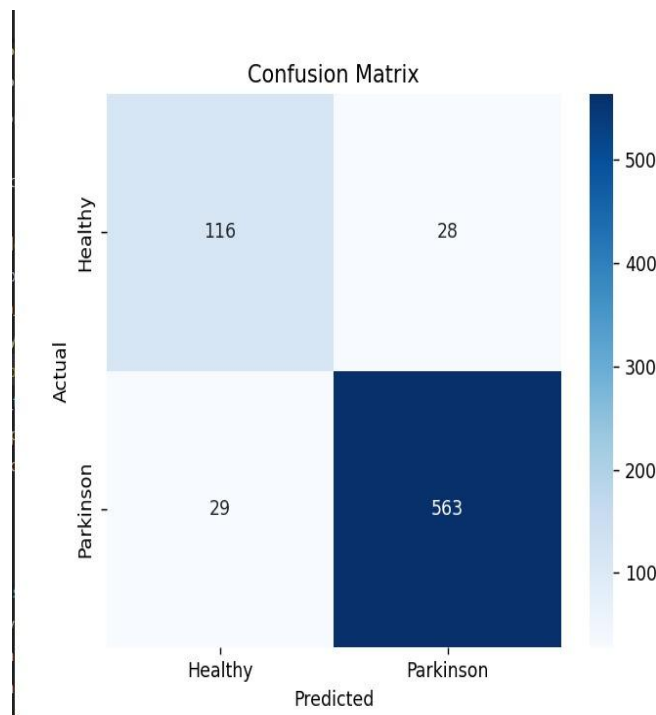


Fig 5.3.4 Confusion Matrix Analysis

This table is used to assess the performance of a classification model. It is a comparison of actual and predicted class labels in a more detailed manner. Information regarding all components of classifier performance is given. The confusion matrix has four elements: TP (True Positive), TN (True Negative), FP (False Positive) and FN (False Negative).

The confusion matrix is where you derive the performance measures.

5.3.4 Grad-CAM Visualisation Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) is an Explainable AI (XAI) technique that highlights the regions in an image responsible for a prediction. Grad-CAM visualises the relevant areas of handwriting that contribute to the model prediction of PD,

which include shaking tremors, unsteady curves, and trembling strokes. Red/yellow are the highly relevant parts of the image and blue are the low importance parts.

Metric	Value
Accuracy	92.26%
Precision (Macro AVG)	87.63
Recall (Macro AVG)	87.83
F1-Score (Macro AVG)	87.73
AUC (ROC)	95.48

Table 5.3.5. Proposed System Accuracy Table

This visualisation gives the user a clear understanding of why the model makes its decision and increase the system transparency, trust, interpretability and reliability suitable for health care uses.

5.4 Challenges and Limitations

5.4.1 Challenges

There were numerous challenging aspects in the implementation and testing phase of the proposed Parkinson's Disease detection system. The major problem was the availability of a good quality database of handwriting. There were problems like differences in handwriting (styles), image quality, writing instrument used and drawing pattern, thus making the classification difficult. Another problem was the class imbalance in the data, the data with Parkinson's had a much larger or smaller amount than that with healthy individuals, making the predictions less reliable and the performance metrics biased. The training of hybrid CNN-LSTM consumed a lot of processing time and a high computation. Choosing appropriate hyperparameters (e.g., learning rate, batch size and sequence length) was hard. For a medical application, an interpretable model is very important, because decisions can be traced and improved further. The use of Grad-CAM for explainability was a good step to understand the model, but it also required a good amount of work in implementation and testing. Finally the proposed system was able to understand good handwriting patterns and give reliable results.

5.4.2 Limitations

While the proposed Hybrid CNN-LSTM model demonstrated effective classification, there are some limitations. The model takes only handwriting and spiral images to diagnose Parkinson's Disease. No additional clinical information of the patient like history, voice features or medical tests is used in the model. Accuracy depends greatly on the quality and quantity of training data. A small data set may lack generalization for test images. The performance of the system might be sensitive to handwriting patterns, image resolution and noise level of images. Lastly, the prediction produced by the model is AI based, so it cannot be used as substitute for clinical diagnosis. The heatmaps produced by Grad-CAM can increase interpretability but are not always giving complete clinical insights. Future work may include using large scale datasets, multimodal data and advanced deep learning techniques.

6. CONCLUSION & FUTURE WORK

Conclusion

In this paper, we propose an intelligent explainable method for the early detection of Parkinson's disease using handwriting and spiral images. A Hybrid CNN-LSTM model was developed for the extraction of spatial and sequential patterns from the handwriting data. A Convolutional Neural Network (CNN) was used to extract important visual features such as tremor, uneven strokes, instability of handwriting. Meanwhile, the Long Short-Term Memory (LSTM) network was trained to learn the sequential correlation between the image patches. These two deep learning networks combined improved classification performance.

To inject transparency and trustworthiness into the framework, Explainable Artificial Intelligence (XAI) utilizing Grad-CAM was employed. The Grad-CAM created heatmaps of important areas in handwriting images, showing the areas the CNN-LSTM model concentrated on to predict the classification output. In this case, the users would be able to understand the decision-making of the model and increase the confidence of the prediction result.

The experimental results showed that the proposed system achieved a promising classification performance with accuracy 92.26%, precision 0.8763, recall 0.8783, F1-score 0.8773 and AUC score 0.9548. These scores reflect the capacity of the Hybrid CNN-LSTM model to discriminate between Healthy handwriting sample and Parkinson's disease handwriting sample.

In conclusion, the proposed system provides an efficient, reliable and human friendly method for Parkinson's disease detection from handwriting. The proposed framework can help medical practitioners in early diagnosis through the combination of deep learning models and XAI techniques. To improve the accuracy and usability, some potential directions are to use a larger dataset, integrate with multimodal medical data, and design novel deep learning structures.

Future Work

The suggested Hybrid CNN-LSTM model for PD detection has exhibited some striking results in effectively classifying handwritten images into healthy and Parkinson's classes. The developed system showed high accuracy and reliable performance. However, many future works can be done to increase model performance and for real-world health care applications. Another way to improve this is to employ multimodal data sources such as speech signals, gait analysis data, biomedical sensor signals and brain imaging data. Integrating more than one information sources may help the model to learn more disease-related features, and hence improve the prediction accuracy.

A larger and more diverse dataset can be gathered from different hospitals and medical institutions in different geographical locations. A huge dataset would make the model robust, generalisable and scaleable. More advance DL models such as Vision Transformers (ViT), EfficientNet, attention networks, transformer based hybrid models and the like could be explored for better feature extraction and classification performance. Data augmentation and transfer learning techniques can be helpful to handle the data shortage problems and to improve the model performance.

Another future work could be the development of real-time diagnosis systems through web and mobile applications, where the professional healthcare providers could use these applications to pre-screen for Parkinson's Disease rapidly and economically. Furthermore, explanations-based AI methods can be further applied to increase confidence in model predictions. Besides classification of the images into healthy and PD patients, disease severity assessment and progression tracking can also be the area of future research using longitudinal patient data.

7. REFERENCES

- [1] U. B. Balci, A. G. Yildirim, and M. S. Aslan, “Parkinson’s Disease Detection from Spiral and Wave Drawings Using Deep Learning,” *IEEE Conference on Signal Processing and Communications Applications*, 2019.
- [2] R. Sharma, S. Kumar, and P. Singh, “Automated Detection of Parkinson’s Disease Using Deep CNN,” *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 5, pp. 112–118, 2020.
- [3] S. Sivaranjini and C. M. Sujatha, “Deep Learning Based Diagnosis of Parkinson’s Disease Using Convolutional Neural Network,” *Multimedia Tools and Applications*, vol. 79, pp. 15467–15479, 2019.
- [4] A. Grover, N. Gupta, and R. Sharma, “CNN-Based Parkinson’s Detection Using Voice and Image Data,” *Journal of Medical Systems*, vol. 45, no. 3, pp. 1–10, 2021.
- [5] Y. Zhang, H. Wang, and X. Liu, “Hybrid CNN-LSTM Model for Parkinson’s Disease Diagnosis,” *IEEE Access*, vol. 9, pp. 112345–112356, 2021.
- [6] M. Rahman, S. Ahmed, and T. Hossain, “Parkinson’s Detection Using CNN-LSTM on Biomedical Signals,” *Biomedical Signal Processing and Control*, vol. 74, pp. 103524, 2022.
- [7] A. Khan, M. Ali, and S. Hassan, “Multimodal CNN-LSTM Framework for Parkinson’s Prediction,” *Expert Systems with Applications*, vol. 213, pp. 118921, 2023.
- [8] M. Little, P. McSharry, E. Hunter, J. Spielman, and L. Ramig, “Suitability of Dysphonia Measurements for Telemonitoring of Parkinson’s Disease,” *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 4, pp. 1015–1022, 2009.
- [9] B. E. Sakar, M. E. Isenkul, C. O. Sakar, A. Sertbas, F. Gurgun, S. Delil, H. Apaydin, and O. Kursun, “A Comparative Analysis of Speech Signal Processing Algorithms for Parkinson’s Disease Classification and the Use of the Tunable Q-Factor Wavelet Transform,” *Applied Soft Computing*, vol. 13, no. 4, pp. 1968–1979, 2013.
- [10] A. Tsanas, M. Little, P. McSharry, and L. Ramig, “Novel Speech Signal Processing Algorithms for High-Accuracy Classification of Parkinson’s Disease,” *IEEE Transactions on Biomedical Engineering*, vol. 59, no. 5, pp. 1264–1271, 2012.

- [11] R. Prashanth, S. Roy, P. K. Mandal, and S. Ghosh, “High-Accuracy Detection of Parkinson’s Disease Through Multimodal Features and Machine Learning,” *International Journal of Medical Informatics*, vol. 90, pp. 13–21, 2016.
- [12] C. Salvatore, A. Cerasa, and I. Castiglioni, “Machine Learning on Brain MRI Data for Differential Diagnosis of Parkinson’s Disease,” *Journal of Neuroscience Methods*, vol. 222, pp. 230–237, 2014.
- [13] P. Drotar, J. Mekyska, M. Rektorova, Z. Smekal, and I. Faundez-Zanuy, “Evaluation of Handwriting Kinematics and Pressure for Differential Diagnosis of Parkinson’s Disease,” *Artificial Intelligence in Medicine*, vol. 67, pp. 39–46, 2016.
- [14] C. Pereira, D. Pereira, A. Silva, and J. Cardoso, “Handwritten Dynamics Assessment for Parkinson’s Disease Detection Using Machine Learning,” *Expert Systems with Applications*, vol. 63, pp. 308–316, 2016.
- [15] B. Eskofier, S. Lee, and J. Daneault, “Recent Machine Learning Advancements in Sensor-Based Mobility Analysis for Parkinson’s Disease,” *Current Neurology and Neuroscience Reports*, vol. 16, no. 10, pp. 1–8, 2016.